

portfolio.

D W I K I
R A H M A N

2025

SELECTED PROJECT

-SOLIDWORKS 3D MODELING
-POWER BI DATA VISUALIZATION

About me



D W I K I
R A H M A N

Certified AK3U (General Safety Expert) |
Mechanical Engineering | 3D CAD &
SolidWorks Enthusiast | TOEFL ITP 523 |
Continuous Learner

Mechanical Engineering graduate from
Universitas Andalas with academic focus on
composite materials, especially TiO_2 -
Fiberglass Epoxy composites through thesis
project. Hands-on industrial experience from
an internship at PT Semen Padang, focusing on
conveyor belt systems and safety practices.

Certified Calon Ahli K3 Umum (AK3U) by the
Ministry of Manpower (Kemnaker RI) and holder
of TOEFL ITP score 523, supporting collaboration
in diverse and professional environments.

Passionate about Engineering, Safety, 3D
design, composites, and continuous learning.

Open to all opportunities in engineering
design, safety, and innovation.

Contact Me

✉ dwiki15rahman@gmail.com

☎ (+62) 813 6863 5794

🌐 <https://www.linkedin.com/in/dwikirahmanst>

🐙 <https://github.com/dwikirahman/Project-and-Practice>

Background & Qualifications



Education and Organization



BACHELOR DEGREE OF MECHANICAL ENGINEERING (2023)

GPA 3.20/4.00

UNIVERSITAS ANDALAS

PADANG, INDONESIA



HIMPUNAN MAHASISWA MESIN UNIVERSITAS ANDALAS (2018)

UNIVERSITAS ANDALAS

PADANG, INDONESIA

THESIS

The Effect of TiO_2 Filler on Glass Fiber-Epoxy Composites Toward Density and Mechanical Properties: Tensile and Compressive Strength.

This test is intended to determine the effect of Titanium Dioxide (TiO_2) filler on the density and mechanical properties, specifically the tensile and compressive strength, of glass fiber epoxy composites.

LINK TO THESIS



Internship Experience



PT. SEMEN PADANG

Service Contract Division of Maintenance Inspection Unit and PGOH (Jan 2022)

PADANG, INDONESIA

LINK TO CERTIFICATE



CASE STUDY

Hot and Cold Splicing Techniques on Conveyor Belts by the Contract Service Team – PT. Semen Padang

- To understand the belt conveyor system, including its components and their respective functions as used in the company.
- To understand the belt conveyor splicing process (hot and cold splicing) as implemented in the company.

Certification

LINK TO CERTIFICATE



The Ministry of Manpower of the Republic of Indonesia

OCCUPATIONAL SAFETY AND HEALTH EXPERT CANDIDATE

Nov 2024

JAKARTA, INDONESIA



Educational Testing Service

TOEFL ITP 523

March 2024-March 2026

PALEMBANG, INDONESIA



Skills and Tools

Soft Skills: Communication, Teamwork, Problem Solving, Continuous Learning

Technical Skills: HSE, 3D Modeling, Data Visualization, English Communication

Tools: SolidWorks, AutoCAD, Microsoft Office, Power BI, Canva, Adobe Photoshop

SOLIDWORKS PROJECT

Tools: SolidWorks 2024
(Competent), SolidWorks Visualize
2024 (Novice)

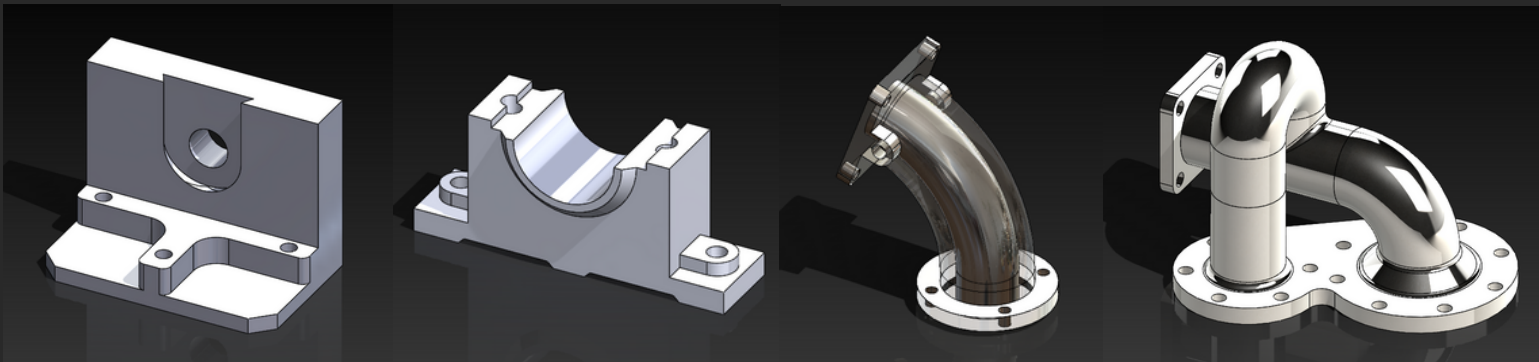


Course: PROMEKANIKA SolidWorks
Bootcamp – Design Engineering
(ongoing, 2025), preparation for
CSWA & CSWP Certification

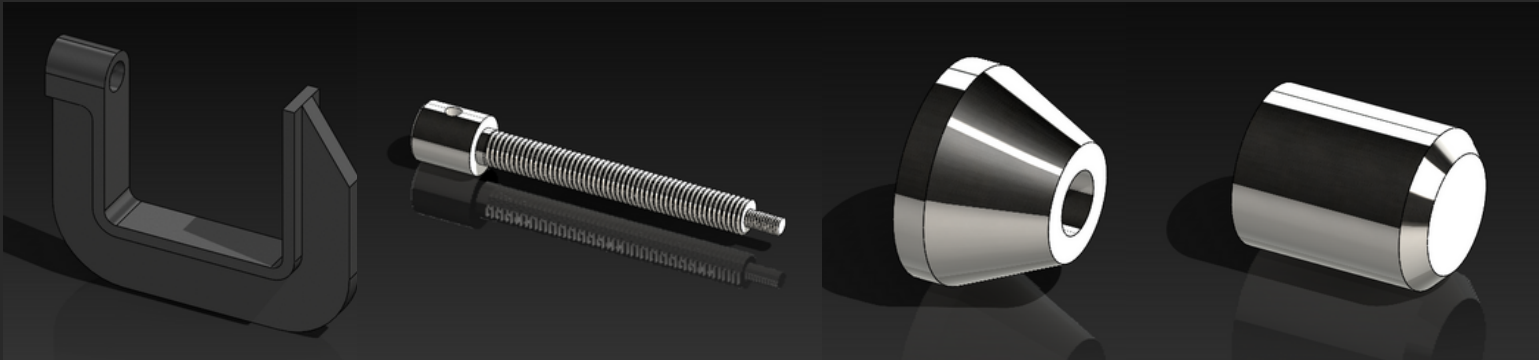


Boot Camp Design Engineering

SolidWorks 3D Model-Part Drawing



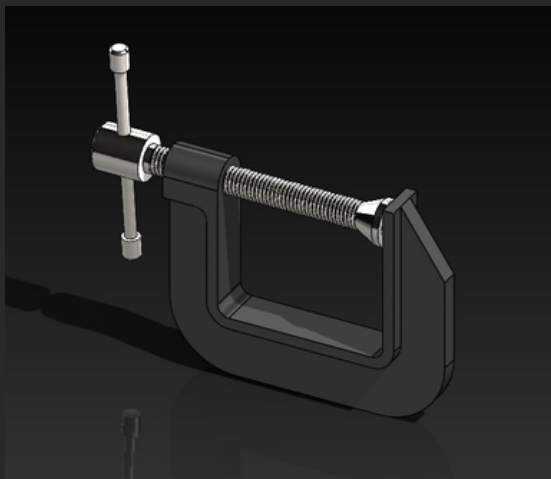
Created several 3D models from advanced technical drawings provided during the course, Developed skills in sketching and applying features such as extrude, revolve, sweep, cut, and more.



Created several parts for assembly while learning features for detailed drawings such as holes, threads, and more.

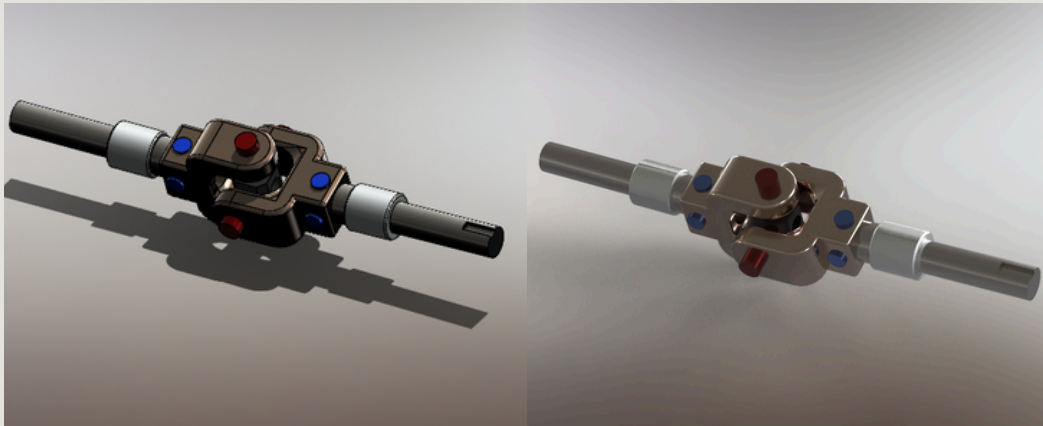
SolidWorks 3D Model-Part Assembly

Assembled multiple created parts into unified components by applying features such as mates, mirror, preferences, and others.



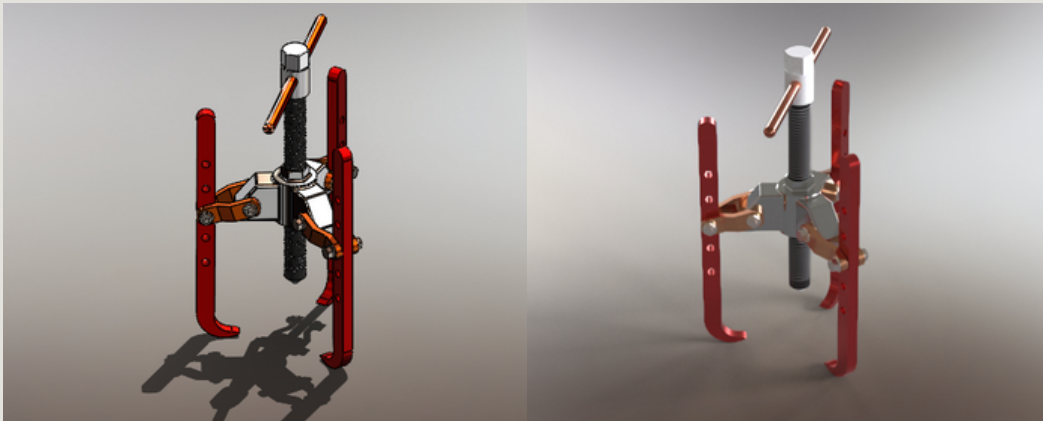
Self-Project

Redrew several components found across the internet for learning purposes and created rendered images.



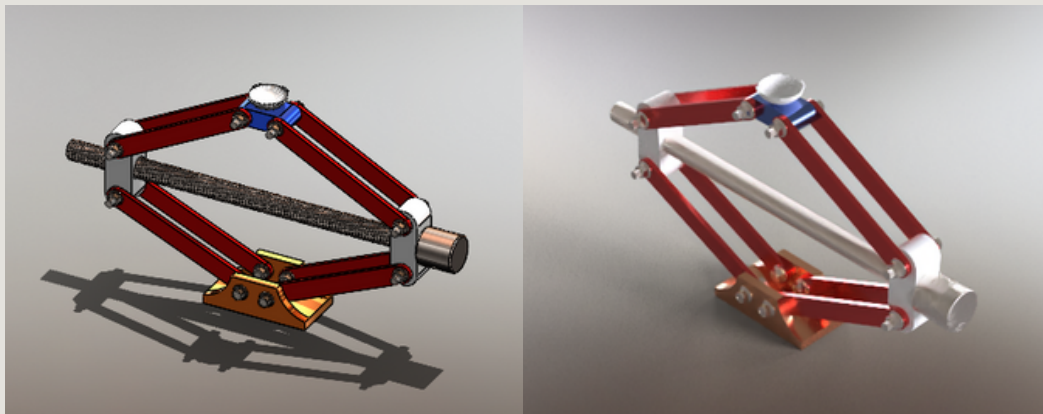
Universal Joint In Assembly

The assembly is made up of 6 parts.



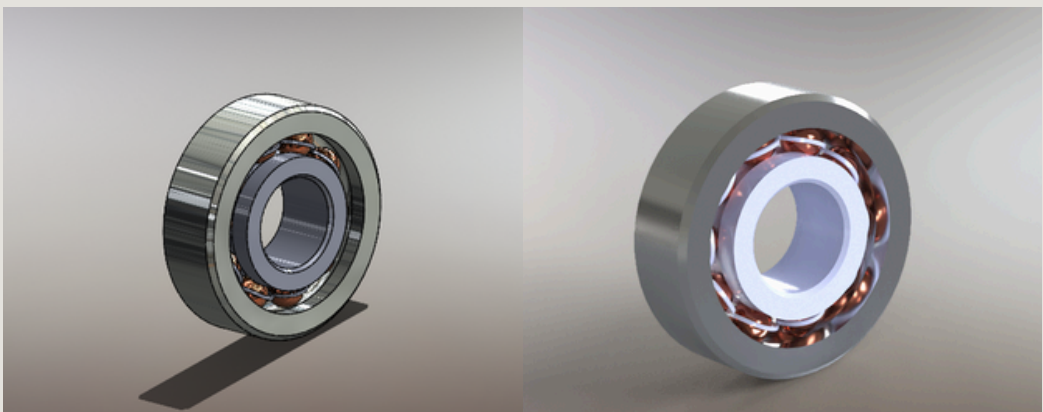
3D Tracker Assembly

The assembly is made up of 5 main parts along with nuts and bolts from the toolbox feature.



3D Scissor Jack Assembly

The assembly is made up of 6 main parts, along with nuts and bolts from the toolbox feature, as well as several mirrored duplicate components.

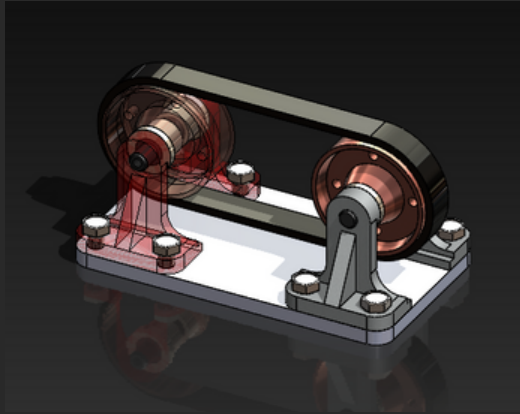


Ball Bearing Assembly

The assembly is made up of 6 main parts, along with bearing pins from the toolbox feature, 12 Balls, as well as several mirrored duplicate components.

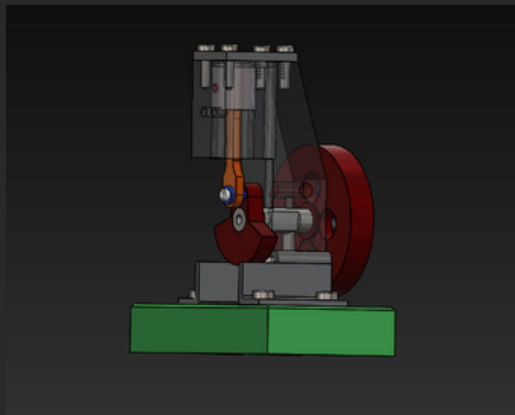
Self-Project

Redrew several components found across the internet for learning purposes and created rendered images.



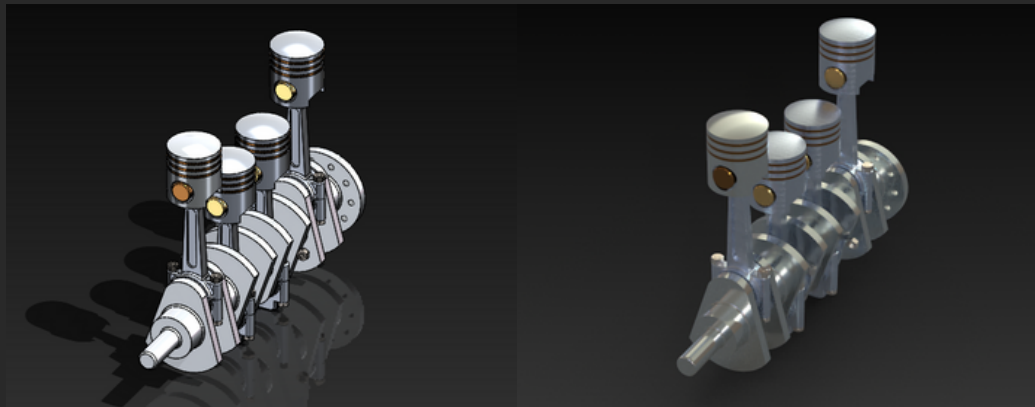
Belt Drive Mechanism Assembly

The assembly is made up of two rollers combined with a belt to create the mechanism. Each roller consists of five parts, along with nuts and bolts from the toolbox feature.



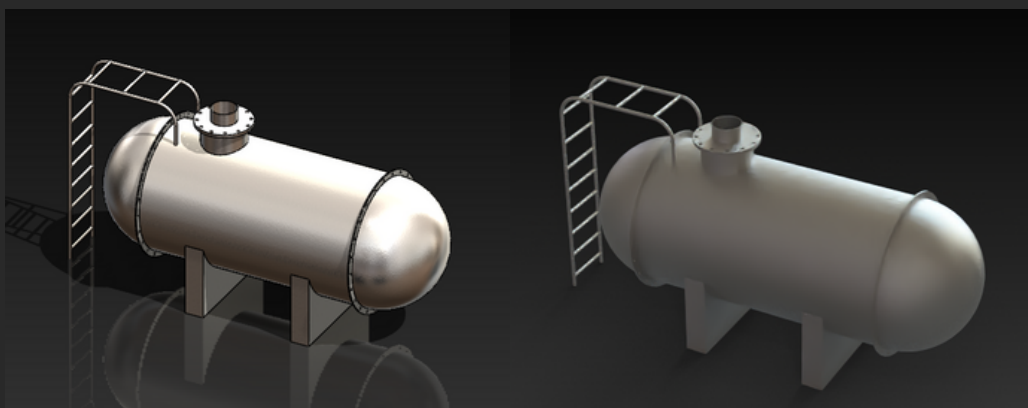
Pneumatic Motor Assembly

The assembly is made up of twelve parts including washer and pins along with the nuts and bolts from toolbox feature.



Cylinder Engine Mechanism Assembly

The assembly is made up of six parts, using features such as Mirror to duplicate the caps into four.

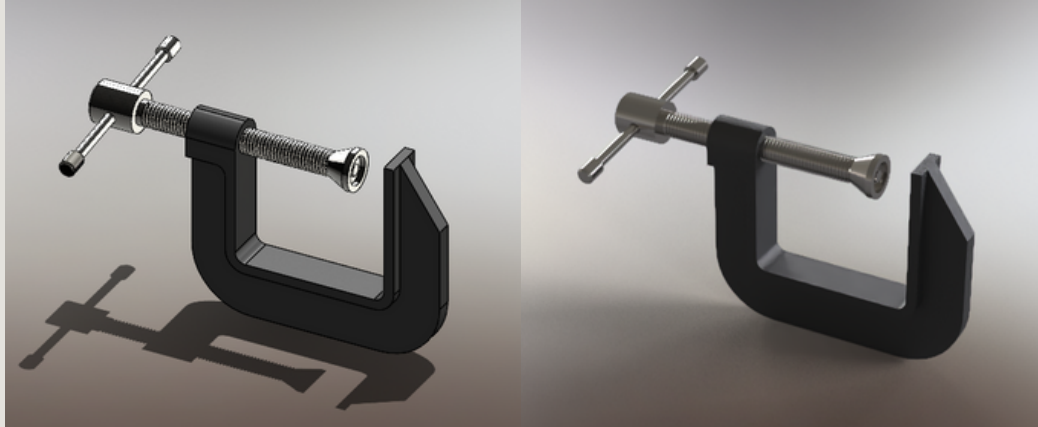


Water Tank 3D Drawing

The drawing was sketched directly in SolidWorks Part, using several features such as Swept for the ladder.

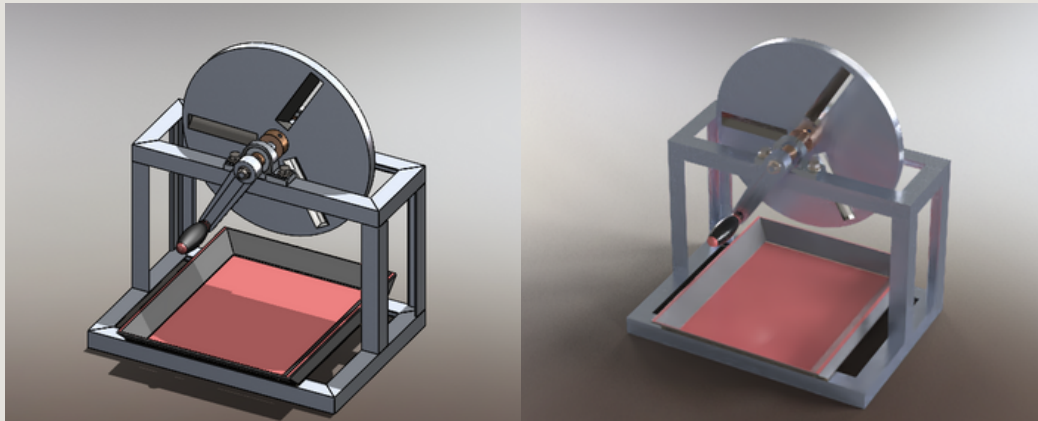
Self-Project

Redrew several components found across the internet for learning purposes and created rendered images.



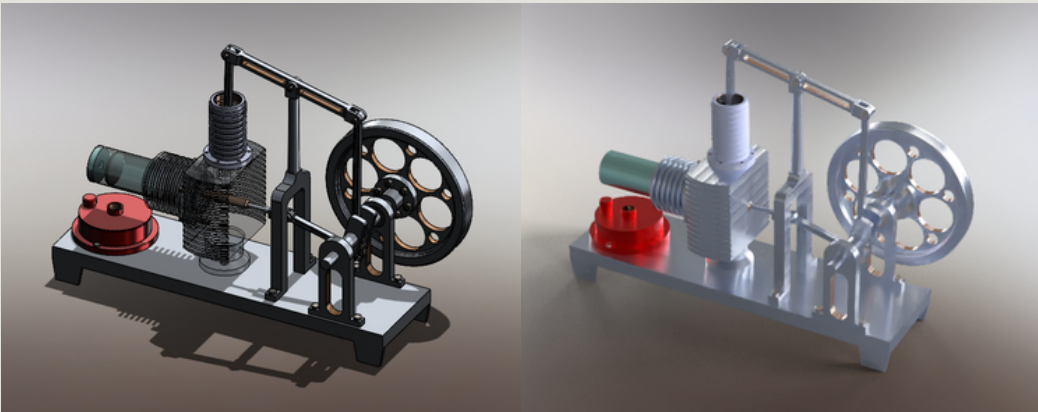
G Clamp Assembly

The assembly is made up of five parts. The parts were created using several features such as Hole and Thread.



Manual Slicer Machine Assembly

The assembly is made up of nine main parts, along with nuts and bolts from the Toolbox feature. This slicer is designed for small items such as cassava or other fruits of similar size.



Stirling Engine Scotch Yoke Type Assembly

The assembly is made up of 35 parts, along with bearings, nuts, and bolts from the toolbox feature, for a total of 86 components.



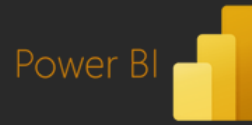
For SolidWorks future projects outside this portfolio, please check my GitHub repository through this link.



POWER BI

DATA VISUALIZATION PROJECT

Tools: Microsoft Power BI (Novice)
Data sources: Open data from the internet and several datasets I created.

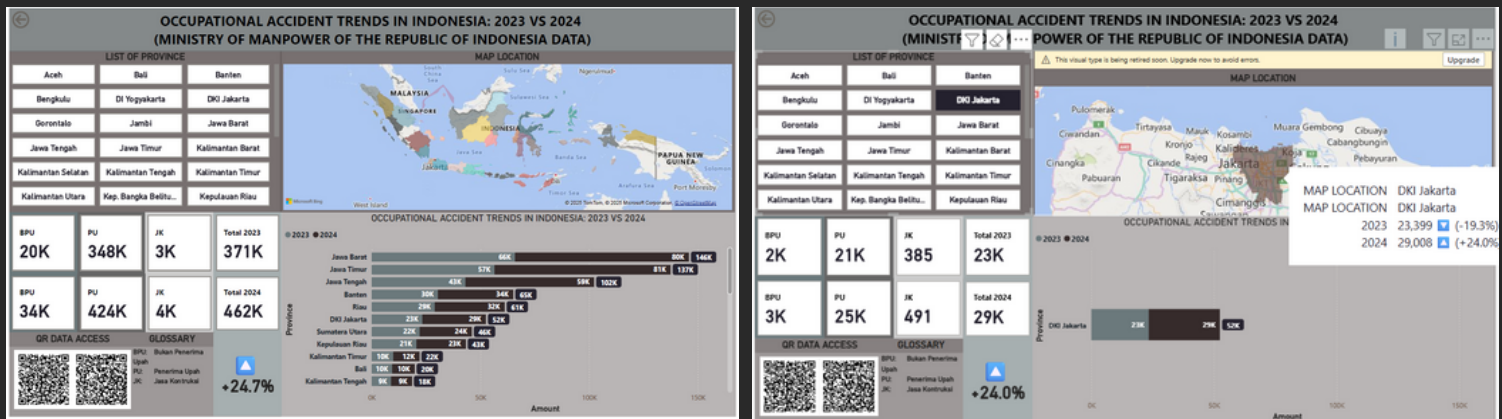


Material Properties of Titanium Dioxide-Glass Fiber Epoxy Composite Dashboard



The data used for this dashboard is based on my thesis research.

Occupational Accident Trends in Indonesia: 2023 vs 2024 Dashboard



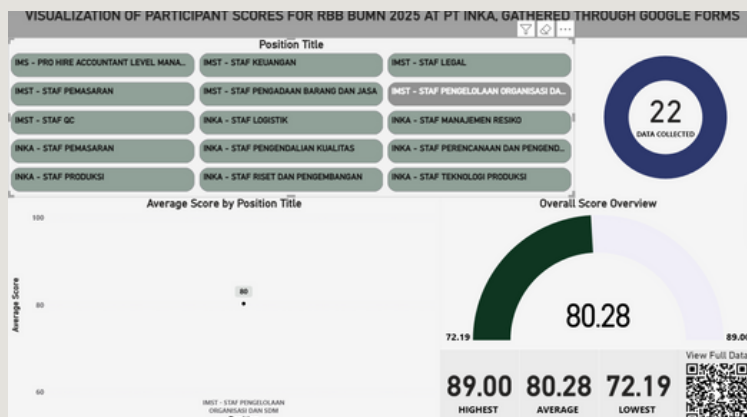
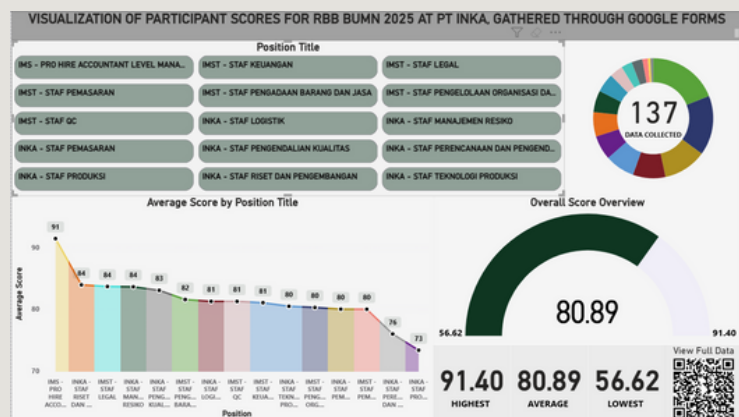
The data used for this dashboard was collected from the open data provided by the Ministry of Manpower of Indonesia.

Field Visit Safety Regulation Findings - PT Budi Nabati Dashboard



The data used for this dashboard is based on my audit findings at the company.

Participant Scores for RBB BUMN 2025 PT INKA Dashboard



The data used for this dashboard was gathered by myself from total of 137 participants using Google Forms



For Power BI Data Visualization future projects outside this portfolio, please check my GitHub repository through this link.



Thank you for taking the time to explore my portfolio. I am always eager to learn, grow, and collaborate on future projects.

If you are interested in connecting or discussing opportunities, feel free to reach me through the contacts below.

- Email: dwiki15rahman@gmail.com
- Phone: (+62) 813 68635794
- LinkedIn: <https://www.linkedin.com/in/dwikirahmanst>
- GitHub: <https://github.com/dwikirahman/Project-and-Practice>



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